



COMPLEX COMPUTING PROBLEM

COURSE: CLOUD COMPUTING & VIRTULIZATION

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CLASS TIMING: SUNDAY(2:30 – 05:30 Pm)

Introduction:

Scholarship management is an important process in educational institutions. Traditional methods often involve paperwork, manual verification, and inefficient communication between students and administrators.

To solve these issues, a web-based Scholarship Management Portal was developed and deployed on AWS cloud infrastructure. The portal enables online scholarship applications while utilizing cloud services for scalability, availability, security, monitoring, and storage.

Objectives

- Develop an online scholarship management system.
- Deploy the application on AWS cloud infrastructure.
- Implement secure database connectivity.
- Provide centralized application management.
- Monitor cloud resources using AWS CloudWatch.
- Store documents using Amazon S3.
- Demonstrate Infrastructure as Code using Terraform.

Literature Review / Related Work:

Traditional Scholarship Systems

Conventional scholarship systems rely heavily on manual documentation, resulting in processing delays and increased administrative workload.

Cloud-Based Education Systems

Modern educational platforms leverage cloud computing to provide:

- High availability
- Scalability
- Reduced operational costs
- Automated backups
- Secure data storage

AWS in Educational Applications

AWS provides various services suitable for educational systems:

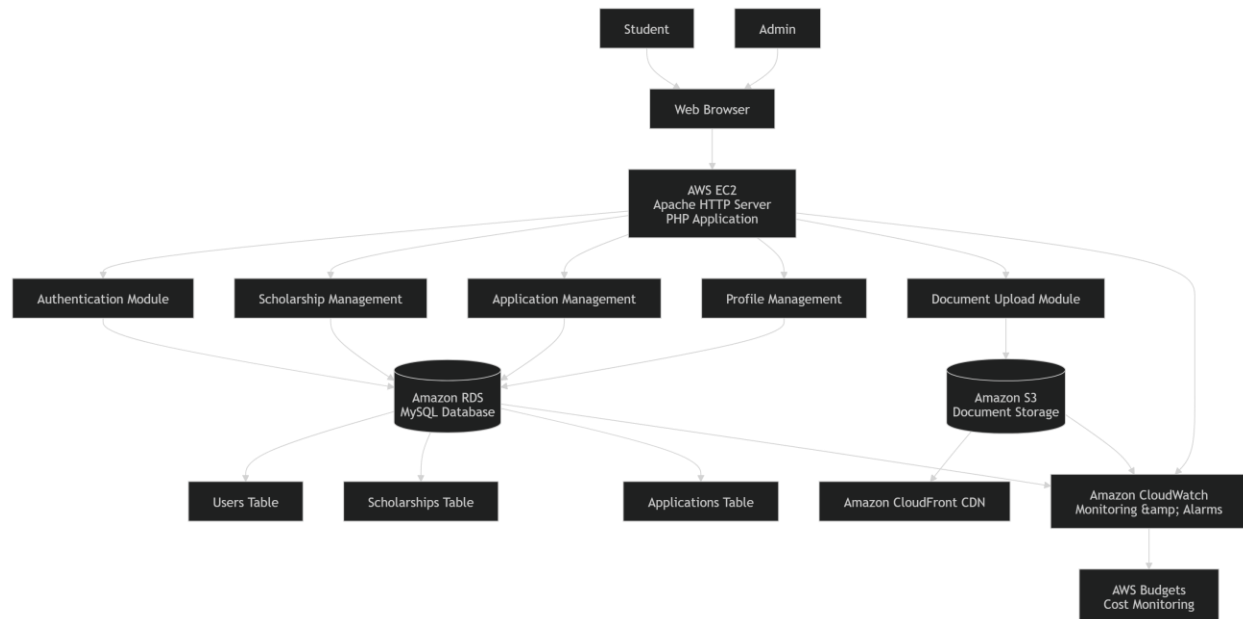
- EC2 for hosting applications
- RDS for managed databases
- S3 for document storage

- CloudFront for content delivery
- CloudWatch for monitoring
- Terraform for infrastructure automation

These services collectively enable the development of reliable cloud-native applications.

System Design and Architecture

Overall Architecture



Architecture Components

1. Student
2. Admin
3. Web Browser
4. ScholarHub Portal (PHP Application)
5. Authentication Module
6. Scholarship Management Module
7. Application Management Module
8. Profile Management Module
9. Document Upload Module
10. MySQL Database
11. Users Table
12. Scholarships Table
13. Applications Table
14. Uploads Folder

AWS Infrastructure:

Virtual Private Cloud (VPC)

A dedicated VPC was configured to isolate project resources and provide secure communication.

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.
☒ Auto-generate

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.
 65,536 IPs
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)

► Encryption settings - optional

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.
☐ 1 ☒ 2 ☐ 3
► Customize AZs

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.
☐ 0 ☒ 2

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.
☐ 0 ☒ 2 ☐ 4
► Customize subnets CIDR blocks

NAT gateways (\$) - **updated** [Info](#)
NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.
☒ None ☐ Regional - new ☐ Zonal

Preview

VPC [Show details](#)
Your AWS virtual network:

ScholarHub-vpc

Subnets (4)
Subnets within this VPC:

eu-north-1a

A

ScholarHub-subnet-public1-eu-

A

ScholarHub-subnet-private1-eu-

eu-north-1b

A

ScholarHub-subnet-public2-eu-

A

ScholarHub-subnet-private2-eu-

vpc-0794b73bb310b4957 / ScholarHub-vpc [Actions](#)

Details [Info](#)

VPC ID

DNS resolution
Enabled

Main network ACL
[acl-0c6ceade7a3888223](#)

IPv6 CIDR (Network border group)
-

Encryption control ID
-

State
☒ Available

Tenancy
default

Default VPC
No

Network Address Usage metrics
Disabled

Encryption control mode
-

Block Public Access
☐ Off

DHCP option set
[dopt-033792d489abccf13](#)

IPv4 CIDR
10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups
-

DNS hostnames
Enabled

Main route table
[rtb-08c2b0a8721dec0ed](#)

IPv6 pool
-

Owner ID

Security Groups

Security Groups were configured to control inbound and outbound traffic.

[EC2](#) > [Security Groups](#) > Create security group

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

Name cannot be edited after creation.

Description Info

VPC Info

vpc-0794b73bb310b4957 (ScholarHub-vpc)

Inbound rules Info

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
SSH	TCP	22	My IP 39.34.189.132/32		Delete
HTTP	TCP	80	Anywhere... 0.0.0.0		Delete
HTTPS	TCP	443	Anywhere... 0.0.0.0		Delete

[Add rule](#)

Rules with source of 0.0.0.0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules Info

Type Info	Protocol Info	Port range Info	Destination Info	Description - optional Info	
All traffic	All	All	Custom 0.0.0.0		Delete

[Add rule](#)

Rules with destination of 0.0.0.0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Ports:

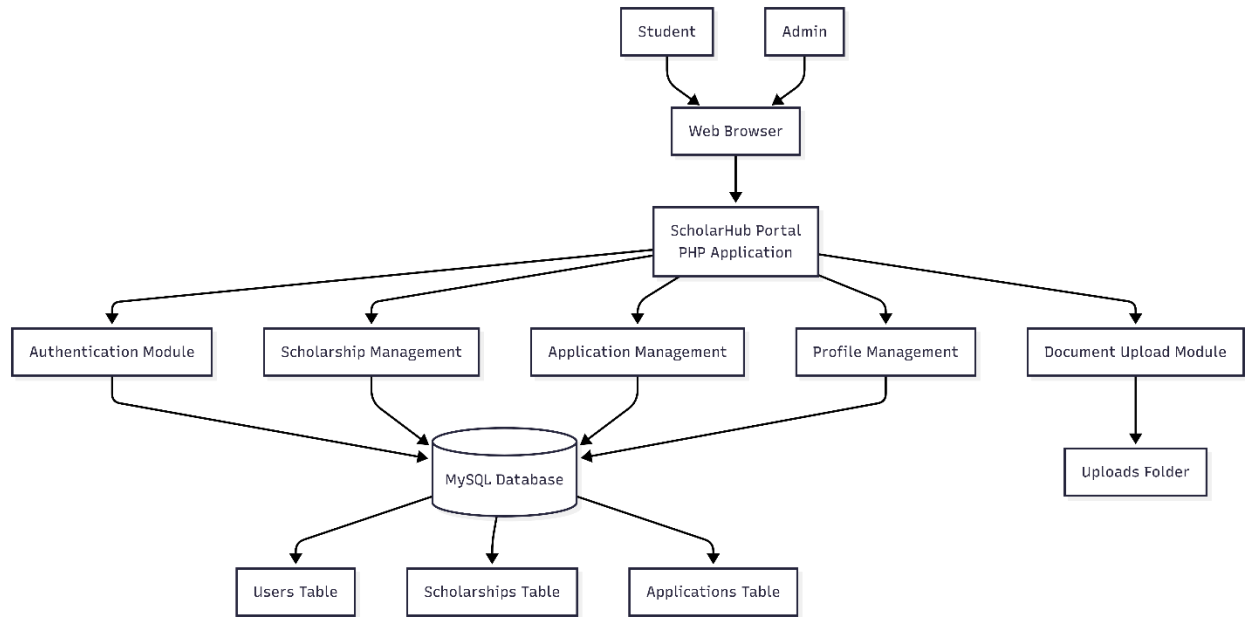
- 22 (SSH)
- 80 (HTTP)
- 443 (HTTPS)
- 3306 (MySQL)

Methodology:

Phase 1: Local Development

The application was initially developed in XAMPP using:

- PHP
- HTML
- CSS
- JavaScript
- MySQL



Database Design:

Database tables:

Users

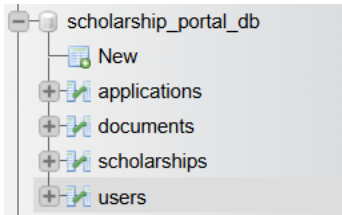
- id
- name
- email
- password
- role

Scholarships

- scholarship_id
- title
- description
- amount

Applications

- application_id
- scholarship_id
- user_id
- document



AWS Deployment:

EC2 Configuration

Amazon EC2 instance was launched and configured as the application server.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-023b6eace47afd3b4 (64-bit (x86), uefi-preferred) / ami-0ca490ac4133fc7e1 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible ▼

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

Free tier eligible ▼

☒ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

[Create new key pair](#)

▼ Network settings

Info

VPC - required

Info

vpc-0794b73bb310b4957 (ScholarHub-vpc)

10.0.0.0/16

↻

Subnet

Info

subnet-0a341061928e45f3e

ScholarHub-subnet-public1-eu-north-1a

↻ Create new subnet ↗

VPC: vpc-0794b73bb310b4957

Owner: 710959681328

Availability Zone: eu-north-1a (eun1-az1)

Zone type: Availability Zone

IP addresses available: 4091

CIDR: 10.0.0.0/20

▼

Auto-assign public IP

Info

Enable

▼

Firewall (security groups)

Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group
 ☒ Select existing security group

Common security groups

Info

Select security groups

▼

↻ Compare security group rules

ScholarHub-Web-SG sg-0e81d94e97a0821e2

✕

VPC: vpc-0794b73bb310b4957

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Configure storage

Info

Advanced

1x

8

GiB

gp3

▼

Root volume, 3000 IOPS, Not encrypted

Add new volume

⌚ Click refresh to view backup information

↻

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

File systems

☐ S3 Files - new
 ☐ EFS

☐ FSx
 ☒ None

SSH Connectivity:

Remote administration was performed through SSH.


```

Directory of C:\Users\MUB\Downloads
06/13/2026  07:39 PM                1,674 ScholarHub-Key.pem
               1 File(s)                1,674 bytes
               0 Dir(s) 16,278,278,144 bytes free

C:\Users\MUB\Downloads>ssh -i "ScholarHub-Key.pem" ec2-user@51.20.184.180
The authenticity of host '51.20.184.180 (51.20.184.180)' can't be established.
ED25519 key fingerprint is SHA256:AxXC3J7swl/yjl0UjJCe8+QNhwwbAVDTitCYmDEx3jY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '51.20.184.180' (ED25519) to the list of known hosts.

#
i\      #####          Amazon Linux 2023
nn\    _#####\
nn\   \###|
nn   \#/
nn    V~'  _-->      https://aws.amazon.com/linux/amazon-linux-2023
    ~~~
      .-.-.-
      /  /  /  /  /
      /m/'  /  /  /

[ec2-user@ip-10-0-2-62 ~]$

```

File Transfer

Project files were transferred from local machine to EC2.

```
C:\Users\MUB>cd C:\xampp\htdocs
C:\xampp\htdocs>scp -i "C:\Users\MUB\Downloads\ScholarHub-Key.pem" -r scholarship_portal ec2-user@51.20.184.180:/home/ec2-user/
add_scholarship.php          100% 3098    18.6KB/s   00:00
approve.php                  100% 207     1.3KB/s   00:00
dashboard.php                100% 3593    21.8KB/s   00:00
edit_scholarship.php         100% 2598    16.3KB/s   00:00
logout.php                   100% 0       0.0KB/s   00:00
manage_scholarships.php      100% 3103    18.4KB/s   00:00
New Text Document.txt        100% 0       0.0KB/s   00:00
reject.php                   100% 207     1.3KB/s   00:00
view_applications.php        100% 2322    12.6KB/s   00:00
view_students.php           100% 1539    9.5KB/s    00:00
db.php                       100% 248     1.5KB/s   00:00
footer.php                   100% 118     0.7KB/s   00:00
header.php                   100% 1661    10.0KB/s   00:00
index.php                    100% 1926    12.0KB/s   00:00
login.php                    100% 2493    15.0KB/s   00:00
logout.php                   100% 118     0.7KB/s   00:00
New Text Document.txt        100% 0       0.0KB/s   00:00
register.php                  100% 2668    16.1KB/s   00:00
Admin Dashboard.jpg          100% 133KB   276.6KB/s  00:00
Apply scholarship Page.jpg    100% 72KB    216.9KB/s  00:00
Avaliable Scholarship Page.jpg 100% 109KB   209.3KB/s  00:00
Amazon RDS Connectivity and Security Configuration.jpg 100% 184KB   539.1KB/s  00:00
Amazon RDS MySQL Connectivity and Security Configuration.jpg 100% 147KB   417.6KB/s  00:00
Amazon RDS MySQL Database Creation Configuration.jpg 100% 136KB   405.8KB/s  00:00
Apache HTTP Server Running on Amazon EC2 Instance.jpg 100% 227KB   644.1KB/s  00:00
database created.jpg         100% 51KB    159.2KB/s  00:00
EC2 to RDS MySQL Connection Successful.jpg 100% 73KB    223.6KB/s  00:00
```

Apache Installation

Apache web server was installed and configured.

```
[ec2-user@ip-10-0-2-62 ~]$ sudo systemctl start httpd
[ec2-user@ip-10-0-2-62 ~]$ sudo systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[ec2-user@ip-10-0-2-62 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-06-13 17:34:17 UTC; 18s ago
     Docs: man:httpd.service(8)
  Main PID: 28508 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 1014)
   Memory: 13.4M
      CPU: 99ms
   CGroup: /system.slice/httpd.service
           └─28508 /usr/sbin/httpd -DFOREGROUND
             └─28560 /usr/sbin/httpd -DFOREGROUND
               └─28564 /usr/sbin/httpd -DFOREGROUND
                 └─28565 /usr/sbin/httpd -DFOREGROUND
                   └─28572 /usr/sbin/httpd -DFOREGROUND

Jun 13 17:34:17 ip-10-0-2-62.eu-north-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Jun 13 17:34:17 ip-10-0-2-62.eu-north-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Jun 13 17:34:17 ip-10-0-2-62.eu-north-1.compute.internal httpd[28508]: Server configured, listening on: port 80
[ec2-user@ip-10-0-2-62 ~]$
```

PHP Installation

PHP runtime environment was installed.

```
[ec2-user@ip-10-0-2-62 ~]$ php -v
PHP 8.5.6 (cli) (built: May 5 2026 21:19:36) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.6, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.6, Copyright (c), by Zend Technologies
[ec2-user@ip-10-0-2-62 ~]$
```

MySQL Client Installation

MySQL client tools were installed for database connectivity.

```
Verifying      : perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Installed:
  mariadb-connector-c-3.3.10-1.amzn2023.0.1.x86_64      mariadb-connector-c-config-3.3.10-1.amzn2023.0.1.noarch
  mariadb105-3:10.5.29-1.amzn2023.0.1.x86_64          mariadb105-common-3:10.5.29-1.amzn2023.0.1.x86_64
  perl-Sys-Hostname-1.23-477.amzn2023.0.9.x86_64

Complete!
[ec2-user@ip-10-0-2-62 ~]$ mysql --version
mysql Ver 15.1 Distrib 10.5.29-MariaDB, for Linux (x86_64) using EditLine wrapper
[ec2-user@ip-10-0-2-62 ~]$
```

Application Deployment:

The ScholarHub application was successfully deployed on an Amazon EC2 instance running Apache HTTP Server and PHP. The application was migrated from the local XAMPP environment to the cloud environment, enabling remote accessibility through a public IP address.

Component	Technology Used
Web Server	Apache HTTP Server
Compute Service	AWS EC2
Database	Amazon RDS MySQL
Storage	Amazon S3
CDN	Amazon CloudFront
Monitoring	AWS CloudWatch
Cost Monitoring	AWS Budgets
IaC	Terraform
Application Language	PHP
Frontend	HTML, CSS, JavaScript

Live Deployment URL

Application URL:

<http://51.20.184.180>

or

<http://ec2-51-20-184-180.eu-north-1.compute.amazonaws.com>

Database and Storage Layer

Amazon RDS

Amazon RDS MySQL was used as the managed database service.

- Automated backups
- Managed maintenance
- Improved reliability
- High availability support

Aurora and RDS > Databases > Create database
?

Create database Info

Free plan has access to limited features and resources
 The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

Engine options

Engine type Info

☐ Aurora (MySQL Compatible)

☐ Aurora (PostgreSQL Compatible)

☒ MySQL

☐ PostgreSQL

☐ MariaDB

☐ Oracle

☐ Microsoft SQL Server

☐ IBM Db2

Connect using

☐ Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☒ Endpoints

Use when connecting through any IDE interface.

Database name

 scholarhub

Master username

 scholarhubadmin

Internet access gateway

☐ Disabled

Port

 3306

Endpoint type

▼ Additional configurations

Connectivity & security

Endpoint

 scholarhub-db.crq4yemco4mt.eu-north-1.rds.amazonaws.com

Port

3306

Networking

Availability Zone

eu-north-1a

VPC

ScholarHub-vpc (vpc-0794b73bb310b4957)

Subnet group

default-vpc-0794b73bb310b4957

Subnets

subnet-05c638486e0cb28d4

subnet-04868f56e8bc2b000

subnet-0a341061928be45f3e

subnet-0c9ff459f8f60fafd

subnet-06859fbcbe2973186

subnet-0830b5ebc214e0771

subnet-0b77d2b66b9c7ee09

Security

VPC security groups

ScholarHub-RDS-SG (sg-04381f018510ed047)

 Active

Publicly accessible

No

Certificate authority

Info

rds-ca-rsa2048-g1

Certificate authority date

May 25, 2061, 02:59 (UTC+05:00)

DB instance certificate expiration date

June 13, 2027, 23:25 (UTC+05:00)

```
[ec2-user@ip-10-0-2-62 ~]$ mysql -h scholarhub-db.crq4yem eo4mt.eu-north-1.rds.amazonaws.com -P 3306 -u scholarhubadmin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]>
```

```

MySQL [(none)]> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| scholarhub |
| sys |
+-----+
5 rows in set (0.020 sec)

MySQL [(none)]> USE scholarhub;
Database changed
MySQL [scholarhub]> SHOW TABLES;
Empty set (0.001 sec)

MySQL [scholarhub]>

```

Databases (1)									
Filter by databases Group resources Modify Actions Create database									
DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...	Size	Recommendation		
scholarhub-db	Available	Instance	MySQL Co...	SECOND	eu-north-1a	db.t4g.micro			

Automated Backups

Automated backup retention was enabled.

Backup

☒ Enable automated backup
Creates a point-in-time snapshot of your database

Please note that automated backups are currently supported for InnoDB storage engine only. If you are using MyISAM, refer to details [here](#).

Backup retention period help
The number of days (1-35) for which automatic backups are kept.

1 day

Backup window help
The daily time range (in UTC) during which RDS takes automated backups.

☒ Choose a window

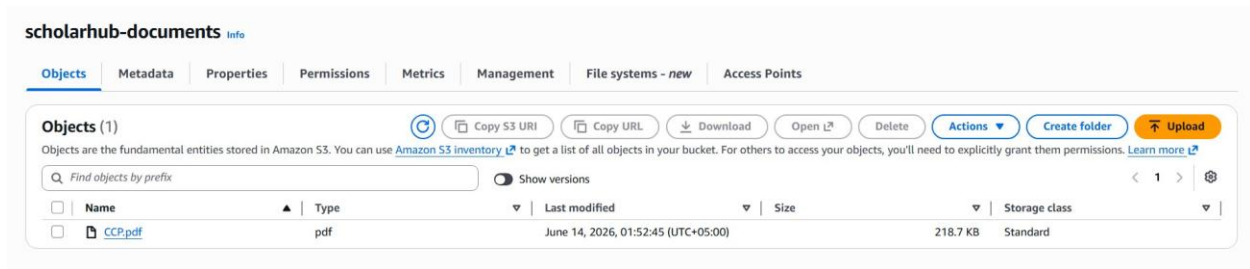
☐ No preference

Amazon S3:

Student supporting documents are stored in Amazon S3.

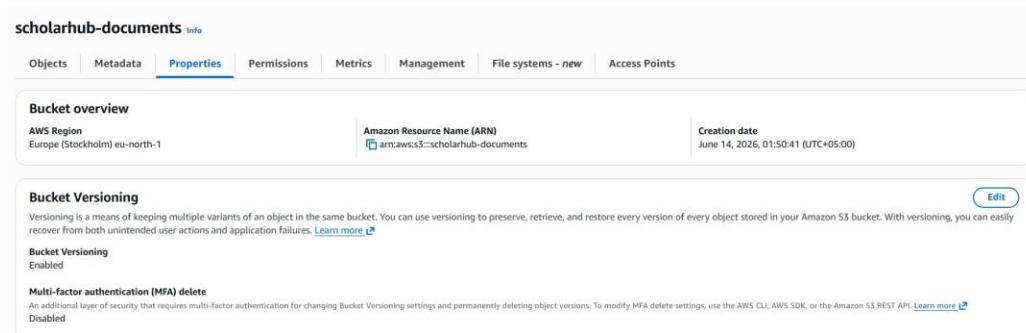
Benefits:

- Durable storage
- Scalability
- Low cost
- Secure file management

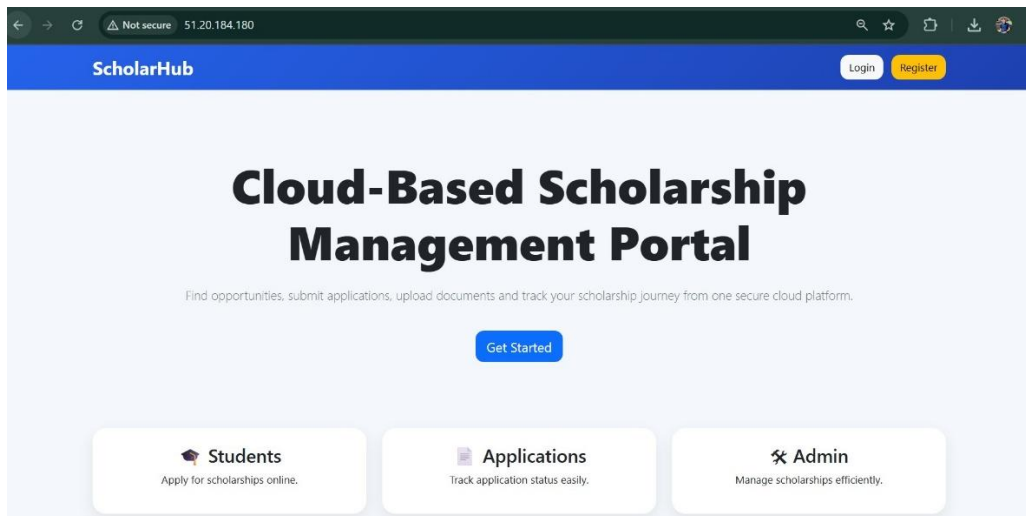


Bucket Versioning:

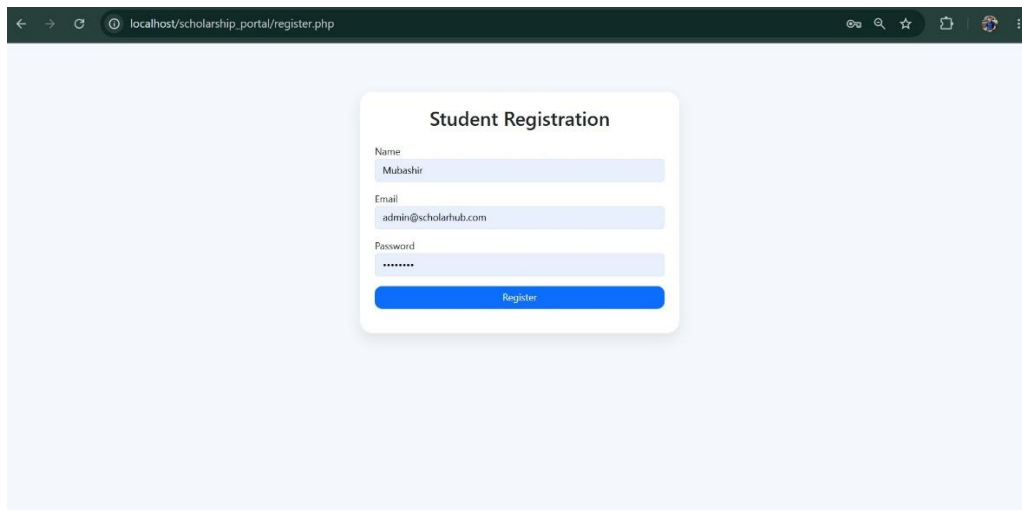
Versioning was enabled to protect against accidental deletion.



Frontend and Application Features



Registration Page:



A screenshot of a web browser showing the 'Student Registration' page. The browser's address bar displays 'localhost/scholarship_portal/register.php'. The registration form is centered on a light blue background. It contains three input fields: 'Name' with the value 'Mubashir', 'Email' with the value 'admin@scholarhub.com', and 'Password' with masked characters '*****'. A blue 'Register' button is positioned at the bottom of the form.

Student Registration

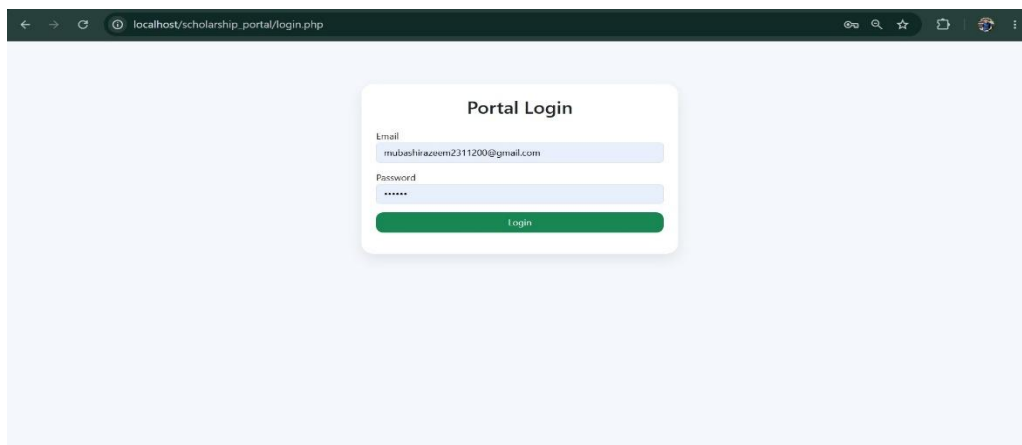
Name
Mubashir

Email
admin@scholarhub.com

Password

Register

Login Page:



A screenshot of a web browser showing the 'Portal Login' page. The browser's address bar displays 'localhost/scholarship_portal/login.php'. The login form is centered on a light blue background. It contains two input fields: 'Email' with the value 'mubashirazem2311200@gmail.com' and 'Password' with masked characters '*****'. A green 'Login' button is positioned at the bottom of the form.

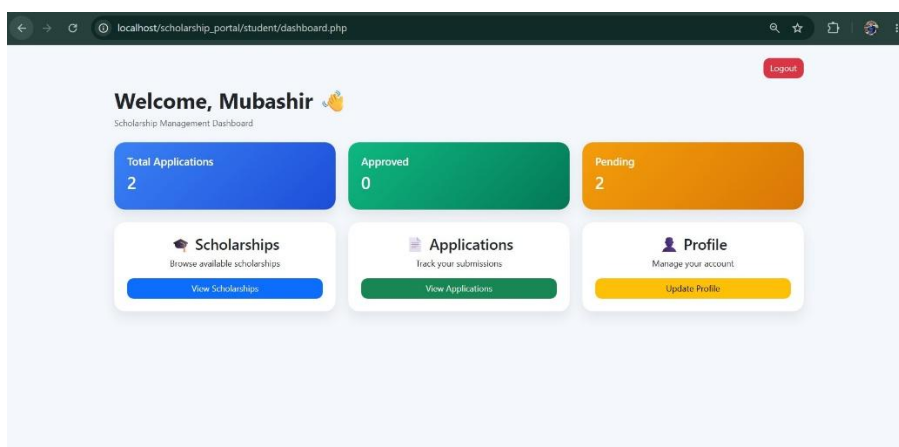
Portal Login

Email
mubashirazem2311200@gmail.com

Password

Login

Student Dashboard:



A screenshot of a web browser showing the 'Student Dashboard' page. The browser's address bar displays 'localhost/scholarship_portal/student/dashboard.php'. The dashboard is titled 'Welcome, Mubashir' with a 'Logout' button in the top right corner. Below the title, there are three summary cards: 'Total Applications' (2), 'Approved' (0), and 'Pending' (2). At the bottom, there are three main sections: 'Scholarships' (Browse available scholarships, View Scholarships button), 'Applications' (Track your submissions, View Applications button), and 'Profile' (Manage your account, Update Profile button).

Welcome, Mubashir 🙌

Scholarship Management Dashboard

Total Applications
2

Approved
0

Pending
2

Scholarships
Browse available scholarships
View Scholarships

Applications
Track your submissions
View Applications

Profile
Manage your account
Update Profile

Logout

Available Scholarships:

Available Scholarships

[Back to Dashboard](#)

Academic Excellence Scholarship
Scholarship Awarded to High Performance Students.
PKR 1200000.00
Deadline: 2026-10-31
[Apply Now](#)

Merit Based
Financial assistance for deserving students.
PKR 2000000.00
Deadline: 2026-03-11
[Apply Now](#)

Apply Scholarship:

Apply For Scholarship

Upload Supporting Document (CNIC / Transcript / Result Card)

[Choose file](#) No file chosen

[Submit Application](#)

[Back to Scholarships](#)

My Applications:

[Back to Dashboard](#)

My Applications

ID	Scholarship	Status	Date
8	Merit Based	Pending	2026-06-13 00:50:05
7	Academic Excellence Scholarship	Pending	2026-06-13 00:49:53

Profile Page:

The screenshot shows a web browser at the URL `localhost/scholarship_portal/student/profile.php`. The page has a blue header with the text "Student Profile". Below the header, there is a link "Back to Dashboard". The main content area contains two forms. The first form is titled "Update Profile" and has two input fields: "Name" with the value "Mubashir" and "Email" with the value "mubashirazeem2311200@gmail.com". Below these fields is a blue button labeled "Update Profile". The second form is titled "Change Password" and has a single input field labeled "New Password". Below this field is a blue button labeled "Change Password".

Admin Dashboard:

The screenshot shows a web browser at the URL `localhost/scholarship_portal/admin/dashboard.php`. The page has a blue header with the text "ScholarHub Admin" and a "Logout" button. Below the header, there is a welcome message "Welcome, Admin" with a user icon and the text "Scholarship Management Administration Panel". The main content area features three large colored boxes representing statistics: "Total Scholarships" with the value "2" (blue box), "Total Applications" with the value "4" (green box), and "Total Students" with the value "3" (orange box). Below these boxes, there are three white boxes with icons and titles: "Scholarships" (graduation cap icon), "Applications" (document icon), and "Students" (group of people icon). Each box contains a description and a button: "Add and manage scholarships." with "Manage Scholarships" button, "Review student applications." with "View Applications" button, and "Manage registered students." with "View Students" button.

Manage Scholarships:

← → ↻ 📍 localhost/scholarship_portal/admin/manage_scholarships.php 🔍 ☆ 📄 👤 ⋮

ScholarHub Admin

Logout

← Back to Dashboard

+ Add Scholarship

Manage Scholarships

ID	Title	Amount	Deadline	Actions
3	Academic Excellence Scholarship	PKR 120000.00	2026-10-31	EditDelete
2	Merit Based	PKR 200000.00	2026-03-11	EditDelete

View Applications

← → ↻ 📍 localhost/scholarship_portal/admin/view_applications.php 🔍 ☆ 📄 👤 ⋮

← Back to Dashboard

Student Applications

ID	Student	Email	Scholarship	Status	Document	Action
8	Mubashir	mubashirazeem2311200@gmail.com	Merit Based	Pending	View File	ApproveReject
7	Mubashir	mubashirazeem2311200@gmail.com	Academic Excellence Scholarship	Pending	View File	ApproveReject
6	Umer Khan	Umer@gmail.com	Academic Excellence Scholarship	Pending	View File	ApproveReject
3	Ali Azam	Ali0@gmail.com	Academic Excellence Scholarship	Pending	No File	ApproveReject

View Students

← → ↻ 📍 localhost/scholarship_portal/admin/view_students.php 🔍 ☆ 📄 👤 ⋮

← Back to Dashboard

Registered Students

ID	Name	Email	Joined
5	Umer Khan	Umer@gmail.com	2026-06-12 23:41:38
4	Ali Azam	Ali0@gmail.com	2026-06-12 19:26:53
1	Mubashir	mubashirazeem2311200@gmail.com	2026-06-12 15:29:33

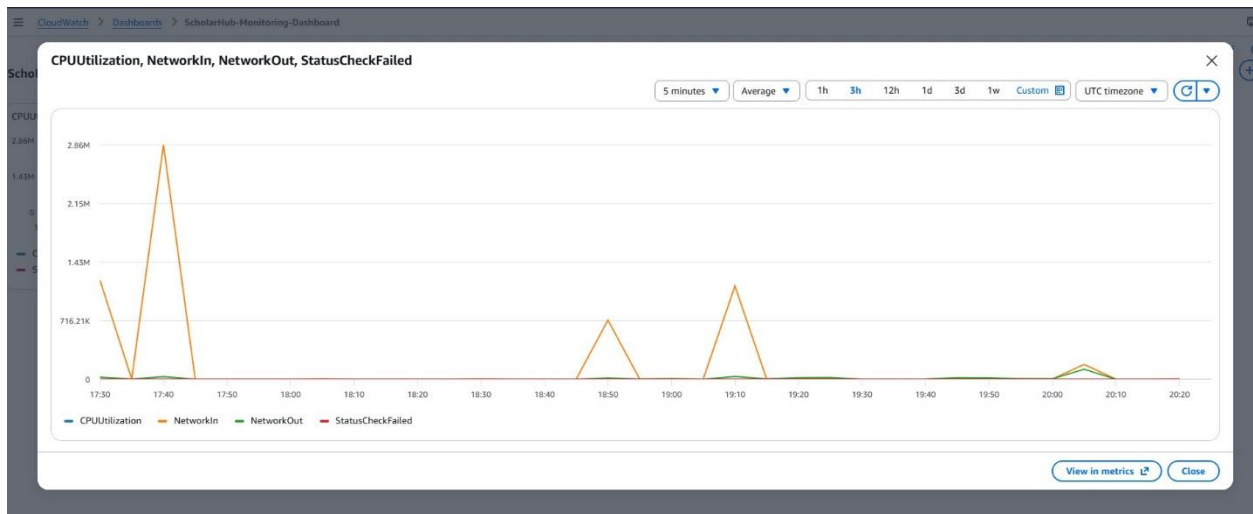
Monitoring and Logging:

AWS CloudWatch

CloudWatch was configured to monitor system health.

Metrics monitored:

- CPU Utilization
- Network In
- Network Out
- Status Checks



CloudWatch Alarm:

An alarm was configured to trigger when CPU usage exceeds 80%.

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ **Create new topic**

☐ **Use topic ARN to notify other accounts**

Send a notification to...

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
mubashirazeem001@gmail.com - [View in SNS Console](#)

[Add notification](#)

Purpose of CloudWatch

CloudWatch continuously collects performance metrics and helps administrators identify issues before they affect application availability.

Successfully created alarm ScholarHub-HighCPU-Alarm. [View alarm](#)

Some subscriptions are pending confirmation
Amazon SNS doesn't send messages to an endpoint until the subscription is confirmed. [View SNS Subscriptions](#)

Alarms | Mute rules - new

Alarms (1)

[Clear selection](#) [Create composite alarm](#) [Actions](#) [Create alarm](#)

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	ScholarHub-HighCPU-Alarm	Insufficient data	Actions enabled Warning	-	2026-06-13 20:44:27	CPUUtiliza minutes

Cost Monitoring:

AWS Budgets was configured.

Budget Limit:

- \$5 Monthly Budget

Benefits:

- Cost visibility
- Spending alerts
- Preventing unexpected charges

✓ Your budget ScholarHub-Budget has been created successfully. After creating a budget, it can take up to 24 hours to populate all of your spend data. [Submit feedback](#) ✕

Budgets (1) [Info](#) [Download CSV](#) [Actions](#) [Create budget](#)

🔍 Find a budget [Type - Show all budgets](#) < 1 > ⚙️

☐	Name	Thresholds	Health status	Billing View	Budget	Amount...
☐	ScholarHub-Budget	✓ OK	✓ Healthy	Primary View	\$5.00	-

CloudFront CDN:

Amazon CloudFront was configured to distribute files stored in S3.

Benefits:

- Faster content delivery
- Global edge caching
- Reduced latency
- Improved user experience

☰ [CloudFront](#) > [Distributions](#) > E2UC5X6Z5HAREX

✓ Successfully created new distribution. ✕

[General](#) [Security](#) [Origins](#) [Behaviors](#) [Error pages](#) [Invalidations](#) [Logging](#) [Tags](#)

Settings [Edit](#)

Name ScholarHub-CDN ✎ Description CloudFront CDN for ScholarHub document delivery from Amazon S3. Price class Use all edge locations (best performance) Supported HTTP versions HTTP/2, HTTP/1.1, HTTP/1.0	Alternate domain names - Add domain	Standard logging Off Cookie logging Off Default root object - Cache tag header -
---	--	---

Infrastructure as Code (Terraform):

Terraform was used to demonstrate Infrastructure as Code (IaC).

- Automated resource creation
- Reproducible infrastructure
- Faster deployment
- Reduced manual configuration

```
C: > xampp > httdocs > scholarship_portal > Terraform > main.tf.txt
1 provider "aws" {
2   region = "eu-north-1"
3 }
4
5 resource "aws_security_group" "scholarhub_sg" {
6   name = "ScholarHub-SG"
7
8   ingress {
9     from_port = 80
10    to_port   = 80
11    protocol = "tcp"
12    cidr_blocks = ["0.0.0.0/0"]
13  }
14
15  ingress {
16    from_port = 22
17    to_port   = 22
18    protocol = "tcp"
19    cidr_blocks = ["0.0.0.0/0"]
20  }
21
22  egress {
23    from_port = 0
24    to_port   = 0
25    protocol = "-1"
26  }
27 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS D:\Projects\Mubashir_RAG_Chatbot> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& d:\Projects\Mubashir_RAG_Chatbot\venv\Scripts\python.exe) .ps1

(venv) PS D:\Projects\Mubashir_RAG_Chatbot>

Results and Discussion:

The system was successfully deployed on AWS and all major functionalities operated correctly.

User Flow 1:

Student Registration to Login to Dashboard

User Flow 2:

Browse Scholarships to Apply to Upload Documents

User Flow 3:

Admin Login to Manage Scholarships to View Applications

Cost Analysis and Security Considerations:

Cost Optimization:

The ScholarHub portal was deployed using AWS Free Tier services to minimize operational costs while maintaining required cloud functionality.

Cost Optimization Techniques

- AWS Free Tier utilized
- Single EC2 instance deployment
- Amazon RDS MySQL Free Tier instance
- AWS Budget alerts configured

- CloudWatch monitoring enabled
- CloudFront caching enabled
- Amazon S3 used for low-cost storage

AWS Budget Monitoring

AWS Budgets was configured to monitor monthly cloud spending and generate alerts when the budget threshold is reached.

✔ Your budget **ScholarHub-Budget** has been created successfully. After creating a budget, it can take up to 24 hours to populate all of your spend data. [Submit feedback](#) ✕

Budgets (1) [Info](#) [Download CSV](#) [Actions](#) [Create budget](#)

🔍 Find a budget Type - Show all budgets < 1 > ⚙️

☐	Name	Thresholds	Health status	Billing View	Budget	Amount
☐	ScholarHub-Budget	✔ OK	✔ Healthy	Primary View	\$5.00	-

Cloud Monitoring

AWS CloudWatch was configured to monitor system performance including CPU utilization, network traffic, and instance health.

✔ Successfully created alarm **ScholarHub-HighCPU-Alarm**. [View alarm](#) ✕

⚠ Some subscriptions are pending confirmation
Amazon SNS doesn't send messages to an endpoint until the subscription is confirmed. [View SNS Subscriptions](#) ✕

[Alarms](#) [Mute rules - new](#)

Alarms (1) [Filter alarms](#) [Quick filters](#) [Clear selection](#) [Create composite alarm](#) [Actions](#) [Create alarm](#)

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	ScholarHub-HighCPU-Alarm	⚠ Insufficient data	✔ Actions enabled Warning	-	2026-06-13 20:44:27	CPUUtilization minutes

Configure actions

Notification

Alarm state trigger
Define the alarm state that will trigger this action. [Remove](#)

☒ **In alarm**
The metric or expression is outside of the defined threshold.
 ☐ **OK**
The metric or expression is within the defined threshold.
 ☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**
☐ Create new topic
☐ Use topic ARN to notify other accounts

Send a notification to...

🔍 [ScholarHub-CPU-Alerts](#) ✕

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
[mubashirazem001@gmail.com](#) - [View in SNS Console](#)

[Add notification](#)

Conclusion:

The Cloud-Based Scholarship Management Portal was successfully designed, developed, and deployed on AWS. The project demonstrates the practical implementation of cloud computing technologies including Amazon EC2, Amazon RDS, Amazon S3, Amazon CloudFront, Amazon CloudWatch, AWS Budgets, and Terraform. The system provides a secure, scalable, and cost-effective solution for managing scholarship applications while fulfilling modern cloud deployment requirements.

The application was successfully hosted on AWS EC2 and integrated with Amazon RDS MySQL for database management. Additional cloud services such as S3 for document storage, CloudFront for content delivery, CloudWatch for monitoring, AWS Budgets for cost control, and Terraform for Infrastructure as Code (IaC) were incorporated to enhance functionality, performance, security, and maintainability. The project successfully demonstrated real-world cloud deployment practices and achieved all major project objectives.

References:

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3. Amazon S3 Documentation. Available: <https://docs.aws.amazon.com/s3/>
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